

Dual-Layer HAMR Media with Ir Intermediate Layer for Multi-Level Magnetic Recording

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Dual-layer granular L1₀-FePt media with a non-magnetic spacer layer could enable multi-level heat-assisted magnetic recording (HAMR). Selecting an appropriate intermediate layer that provides magnetic isolation between the two magnetic recording layers and supports proper growth of the top FePt layer is non-trivial. In this work, we investigated the chemical ordering, magnetic properties, and microstructure of dual-layer media with an iridium intermediate layer. The Ir intermediate layer facilitates the growth of well-ordered L1₀-FePt in the top half of the grain by flattening the bottom FePt layer. Ir is also immiscible with the grain boundary material (*h*BN), enabling well-separated, columnar grains. The magnetic properties of the dual-layer media reflect a majority fraction of L1₀-FePt in both the top and bottom magnetic recording layers.

Index Terms—Areal density, heat-assisted magnetic recording (HAMR), L1₀-FePt, multi-level magnetic recording.

I. INTRODUCTION

HEAT-ASSISTED magnetic recording (HAMR) has enabled substantial increases in storage density of hard disk drives (HDDs) beyond the limits of conventional perpendicular magnetic recording (PMR) [1]. Increasing areal density capability (ADC) beyond 1 Tb/in² means smaller bits, and consequently, smaller magnetic grains in the magnetic recording media. Maintaining thermal stability at these smaller grain dimensions necessitates the use of materials with high magnetocrystalline anisotropy energies. L1₀-FePt has become the standard high-anisotropy material used in HAMR media, with magnetic anisotropy $K_U \sim 4.5$ MJ/m³ [2]. Difficulties for single-layer L1₀-FePt granular media lie in the switching field distribution and bit transition width due to grain non-uniformity.

Multi-level recording is an alternative strategy for increasing ADC and circumvents the challenges associated with continued miniaturization of magnetic grains in single-layer media [3]. One approach to multi-level recording is a system with two discrete magnetic recording layers separated by a non-magnetic intermediate layer. This system could achieve three distinct logical states (-1, 0, 1) and has been theorized to potentially increase ADC by over 50% [4]. Additionally, the ability to tune the Curie temperature and other magnetic properties of the FePt layers separately may also support independent recording of each layer [3].

Recently, attempts to fabricate dual-layer magnetic recording media structures have been reported, where Ru, Ir, or Pt are used as the non-magnetic intermediate layer [5, 6]. However, these dual-layer media structures suffer from interdiffusion between the intermediate layer and the surrounding FePt. This significantly degrades chemical ordering and magnetic properties of the top L1₀-FePt layer. Facilitating proper growth of the top FePt layer with good L1₀ ordering and crystallographic (001) texture is a key requirement for the intermediate layer. In addition to close lattice matching with FePt, high thermal stability at the deposition temperature and immiscibility with FePt and the grain boundary material are

equally important. In this work, we investigated iridium as a candidate intermediate layer material. Ir has a small lattice mismatch with L1₀-FePt of $\sim 0.8\%$ and does not form thermodynamically stable compounds with Pt.

II. METHODS

Dual-layer magnetic recording media were prepared on MgO (001) substrates via magnetron sputtering. The general structure of all samples was FePt-*h*BN (4.5 nm) / Ir-*h*BN (2 nm) / FePt-*h*BN (4.5 nm). The template deposition process for FePt-*h*BN granular media was outlined by C. Xu et al. [7] All samples were deposited at 725°C, following a thermal flush of the MgO substrate at 800°C for 1 hr.

Scanning transmission electron microscopy (STEM) and energy-dispersive X-ray spectroscopy (EDS) were performed using Thermo Scientific Themis Z S/TEM. HAADF-STEM plane-view images and MIPAR Image Analysis Software were used to measure grain sizes and grain center-to-center pitch distances. Magnetic measurements were performed on the MPMS 3 SQUID-VSM (Quantum Design).

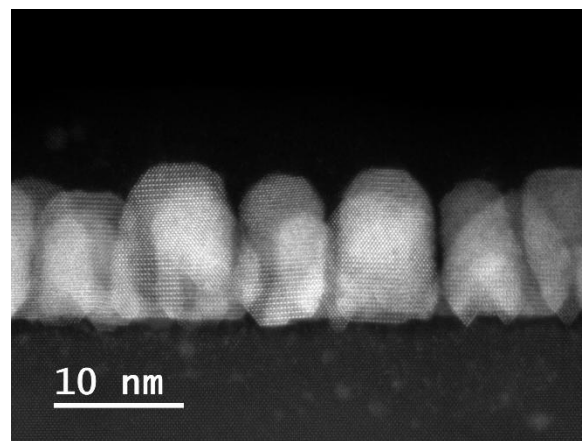


Fig. 1. High-resolution HAADF-STEM of FePt (4.5 nm) / Ir (2 nm) / FePt (4.5 nm) dual-layer media.

III. RESULTS AND DISCUSSION

High-resolution STEM imaging revealed that Ir produces an atomically smooth interface when deposited on FePt. The small lattice mismatch between Ir and $L1_0$ -FePt results in cube-on-cube epitaxy, with the epitaxial relationship $L1_0$ -FePt (001)[100] $\parallel$ Ir (001)[100]. The flat FePt-Ir interface allows for the growth of a second $L1_0$ -FePt layer with the correct (001) texture, as shown in Figure 1. The periodic dark and bright contrast in the micrograph corresponds to alternating planes of Fe and Pt atoms arranged in an $L1_0$ structure. Few regions of disordered FePt were observed in the top magnetic recording layer, which may result from insufficient in-plane tensile strain from the Ir layer to promote the desired structure or texture [6].

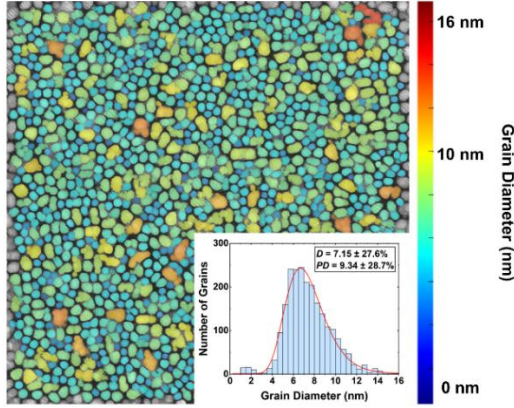


Fig. 2. Plane-view HAADF-STEM of FePt (4.5 nm) / Ir (2 nm) / FePt (4.5 nm) dual-layer media, colored with measured grain diameters. The inset is the grain size distribution fitted to a log-normal distribution to derive an average grain diameter (D) and an average grain-to-grain pitch distance (PD).

While Ogawa et al. observed a network morphology for FePt-C dual-layer media with an Ir spacer layer [6], our samples exhibited granular morphology. Importantly, Ir is immiscible with the grain boundary material (h BN) and yields the desired granular microstructure. The dual-layer media consisted of separated grains with an average grain diameter of $7.2 \text{ nm} \pm 27.6\%$ and an average pitch distance of $9.3 \text{ nm} \pm 28.7\%$. The dual-layer media largely retains the grain size distribution of the single-layer media, with only a slight increase of $\sim 1 \text{ nm}$ in all dimensions.

To characterize potential interdiffusion between the Ir intermediate layer and the adjacent FePt layers, cross-sectional EDS analysis was performed. EDS elemental maps showed minimal diffusion of Ir into either FePt layer. As expected according to the binary alloy phase diagram, Pt does not diffuse substantially into the Ir layer. The diffusion of Fe into the Ir layer was more significant, suggesting that the Ir intermediate layer acts as a diffusion sink in this system. The reduction in Fe content in both recording layers lowers the overall coercive field of the dual-layer media.

The room-temperature M - H loop of the dual-layer media exhibits a shallow inflection in the descending branch at $M/M_s \sim 0.5$ (Figure 3). The step height, less than half the total magnetization reversal, suggests that coercivity in both layers is dominated by well-ordered $L1_0$ -FePt rather than a disordered soft phase. It should be noted that a continuous hysteresis loop

for dual-layer media does not necessarily indicate magnetically coupled recording layers. Nominally decoupled magnetic layers may exhibit a subtle slope change or small kink in the magnetization curve instead of a double hysteresis if the two layers have similar coercive fields or partial overlap in the switching field distributions [8].

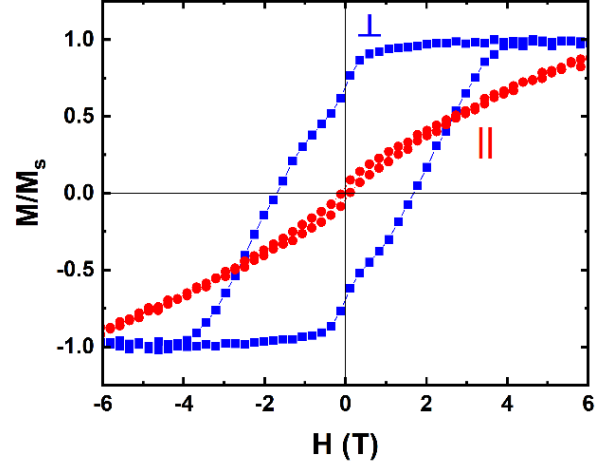


Fig. 3. In-plane ($\parallel$) and out-of-plane ($\perp$) magnetization curves of dual-layer media with a 2 nm Ir spacer layer.

CONCLUSION

We have demonstrated that Ir is a suitable intermediate layer for dual-layer $L1_0$ -FePt granular media. Ir grows epitaxially on the bottom FePt layer and enables the growth of the top FePt layer with good $L1_0$ ordering and texture. Managing Fe migration into the Ir intermediate layer is critical for optimizing the magnetic properties of this system. This can be achieved by introducing elemental dopants in the Ir layer to effectively create a barrier for Fe diffusion. Modifying the $\text{Fe}_x\text{Pt}_{1-x}$ stoichiometry of the magnetic recording layers to compensate for Fe migration into the Ir spacer layer could also be considered.

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